

ANSI Standard (X3.64) Control Sequences
for Video Terminals and Peripherals
with Examples for the IBM PC

Compiled by W.C. Parke, 1986

References:

ANSI X3.64-1979:

Information Systems-Codes, American National Standards
Institute, 1430 Broadway, NY, NY 10018

ISO 6429 standard:

Information Processing-ISO 7-bit and 8-bit Coded
Character Sets-Additional Control Functions for
Character-Imaging Devices, ANSI as above.

Mark L. Siegel, April 1984 BYTE, p. 365

Zenith Programmer's Utility Pack, Ver. 2, 1984

In the table below:

F	is a Final character in an Escape sequence (F from 48 to 126 in the ASCII table) a control sequence (F from 64 to 126).
Fe	is a Final character in ASCII range 64 to 95.
Gs	is a graphic character appearing in strings (Gs ranges from 32 to 126) in the ASCII table.
C0	is the set of 7-bit ASCII control characters from 0 to 31.
C1	the set of control characters available only in 8-bit systems.
Ce set	is a control represented as a single bit combination in the C1 of controls in an 8-bit character set.
I	is an Intermediate character from 32 to 47 (inclusive) in the ASCII table.
P	is a parameter character from 48 to 63 (inclusive) in the ASCII table.
Pn	is a numeric parameter in a control sequence, a string of zero or more characters ranging from 48 to 57 in the ASCII table. Numeric parameters must be in decimal radix.
Ps sequence code	is a variable number of selective parameters in a control with each selective parameter separated from the other by the 59 (which usually represents a semicolon); Ps ranges from 48 to 57 and 59.

Sequence Mnemonic	Sequence Name	Sequence	Default Parameter Value

Introducer sequences:

CSI	Control Sequence Intro	Esc [
SS2	Single Shift 2 (G2 set)	Esc N
SS3	Single Shift 3 (G3 set)	Esc O

Delimiters:

APC	Applicatn Program Command	Esc Fe
DCS	Device Control String	Esc P
OSC	Operating System Command	Esc]
PM	Privacy Message	Esc ^
ST	String Terminator	Esc \

Configuration:

DA	Device Attributes	Esc [Pn c	0
	The VT100 responds to receiving the DA by transmitting the sequence: Esc [? 1 ; Ps c		
DSR	Device Status Report	Esc [Ps n	0
	The console driver will output a CPR sequence.		
STS	Set Transmit State	Esc S	
DMI	Disable Manual Input	Esc \	
EMI	Enable Manual Input	Esc b	
INT	Interrupt	Esc a	
RIS	Reset to Initial State	Esc c	

Editor Functions:

CBT	Cursor Backward Tab	Esc [Pn Z	1
CCH	Cancel Previous Character	Esc T	
CHA	Cursor Horzntal Absolute	Esc [Pn G	1
CHT	Cursor Horizontal Tab	Esc [Pn I	1
CNL	Cursor Next Line	Esc [Pn E	1
CPL	Cursor Preceding Line	Esc [Pn F	1
CPR	Cursor Position Report	Esc [Pn ; Pn R	1, 1
	Report of cursor current line and column. () Do not use as part of MS-DOS Prompt command.)		
CTC	Cursor Tab Control	Esc [Ps W	0
CUB	Cursor Backward	Esc [Pn D	1
	Move cursor back Pn columns. Ignored if cursor in leftmost column.		
CUD	Cursor Down	Esc [Pn B	1
	Move cursor down Pn lines. Ignored if cursor on bottom line.		

CUF	Cursor Forward	Esc [Pn C	1
	Move cursor Pn columns forward. Ignored if cursor at rightmost column.		
CUP	Cursor Position	Esc [Pn ; Pn H	1, 1
	Move cursor to position specified. First parameter is line number and second is column number. No parameters means home position.		
CUU	Cursor Up	Esc [Pn A	1
	Move cursur up Pn lines. Ignored if cursor on top line.		

Sequence Mnemonic	Sequence Name	Sequence	Default Parameter Value
CVT	Cursor Vertical Tab	Esc [Pn Y	
DAQ	Define Area Qualification	Esc [Ps o	0
DCH	Delete Character	Esc [Pn P	1
DL	Delete Line	Esc [Pn M	1
EA	Erase in Area	Esc [Ps O	0
ECH	Erase Character	Esc [Pn X	1
ED	Erase in Display	Esc [Ps J	0
	Erases whole screen and cursor goes to home position.		
EF	Erase in Field	Esc [Ps N	0
EL	Erase in Line	Esc [Ps K	0
	Erases everything between the cursor and the end of the line, including cursor position.		
EPA	End of Protected Area	Esc W	
ESA	End of Selected Area	Esc G	
ICH	Insert Character	Esc [Pn @	1
IL	Insert Line	Esc [Pn L	1
NP	Next Page	Esc [Pn U	1
PP	Preceding Page	Esc [Pn V	1
SD	Scroll Down	Esc [Pn T	1
SL	Scroll Left	Esc [Pn Space @	1
SPI	Spacing Increment	Esc [Pn ; Pn Space G	none
SPA	Start of Protected Area	Esc V	
SR	Scroll Right	Esc [Pn Space A	1
SSA	Start of Selected Area	Esc F	
SU	Scroll Up	Esc [Pn S	1
TBC	Tab Clear	Esc [Ps g	0
TSS	Thin Space Specification	Esc [Pn Space E	none
VPA	Vert Position Absolute	Esc [Pn d	1
VPR	Vert Position Relative	Esc [Pn e	1
VTs	Vertical Tabulation Set	Esc J	

Format Sequences:

FNT	Font Selection	Esc [Pn ; Pn Space D	0, 0
GSM	Graphic Size Modify	Esc [Pn ; Pn Space B	100, 100
GSS	Graphic Size Selection	Esc [Pn Space C	none
HPA	Horz Position Absolute	Esc [Pn `	1
HPR	Horz Position Relative	Esc [Pn a	1
HTJ	Horz Tab w/Justification	Esc I	
HTS	Horizontal Tab Set	Esc H	
HVP	Horz & Vertical Position	Esc [Pn ; Pn f	1, 1
	Moves cursor to line number, column number specified.		
	No parameters moves cursor to home position.		
IND	Index	Esc D	
JFY	Justify	Esc [Ps ; ... ; Ps Space F	0
MC	Media Copy	Esc [Ps i	0

MW	Message Waiting	Esc U	
NEL	Next Line	Esc E	
PLD	Partial Line Down	Esc K	
PLU	Partial Line Up	Esc L	
PU1	Private Use 1	Esc Q	
PU2	Private Use 2	Esc R	
QUAD	Typographic Quadding	Esc [Ps Space H	0
RCP	Restore Cursor Position	ESC [u	
	Restores cursor position saved by SCP.		
REP	Repeat Char or Control	Esc [Pn b	1
RI	Reverse Index	Esc M	

Sequence Mnemonic	Sequence Name	Sequence	Default Parameter Value

RM	Reset Mode	Esc [Ps l	none
	Reset screen width or type. See SM for parameters. Parameter 7 will reset wrap mode (characters past end-of-line are lost.)		
SCP	Save Cursor Position	Esc [s	
	Cursor position will be restored with RCP sequence.		
SEM	Select Edit Extent Mode	Esc [Ps Q	0
SGR	Select Graphic Rendition	Esc [Ps m	0
	Invokes graphic rendition specified by the parameters. All following characters are rendered according to the parameter(s) until the next SGR. Parameter functions on IBM PC:		
	0 All attributes off (normal, white on black)		
	1 Bold on (high intensity)		
	4 Underscore On (monochrome displays only)		
	5 Blink on		
	7 Reverse Video on		
	8 Cancelled on (invisible)		
	30 Black foreground		
	31 Red foreground		
	32 Green foreground		
	33 Yellow foreground		
	34 Blue foreground		
	35 Magenta foreground		
	36 Cyan foreground		
	37 White foreground		
	40 Black foreground		
	41 Red background		
	42 Green background		
	43 Yellow background		
	44 Blue background		
	45 Magenta background		
	46 Cyan background		
	47 White background		
SM	Select Mode	Esc [Ps h	none
	Invokes the screen width or type specified. IBM PC parameters:		
	0 40 x 25 monochrome		
	1 40 x 25 color		
	2 80 x 25 monochrome		
	3 80 x 25 color		
	4 320 x 200 color		
	5 320 x 200 monochrome		
	6 640 x 200 monochrome		
	7 wrap at end of line (word wrap)		

IBM PC Keyboard Reassignment:

Keyboard reassignment uses an undefined ANSI X3.64-1979 code sequence to define a character or set of characters send by the console whenever a given key is pressed on an IBM PC compatible. The control sequence is:

```
ESC [ Pn ; Pn ; ... Pn p
```

or

```
ESC [ "string" ; p
```

The first ASCII code in the sequence defines which code is being mapped. If the first code is 0 (nul), then the second code is the extended IBM keycode. The remaining numbers define the sequence of ASCII codes generated when this key is intercepted, or the string (enclosed in quotes) to be generated. For example:

ESC [65;81p	A becomes Q
ESC [0;68;"dir";13p	F10 generates dir followed by a CR.